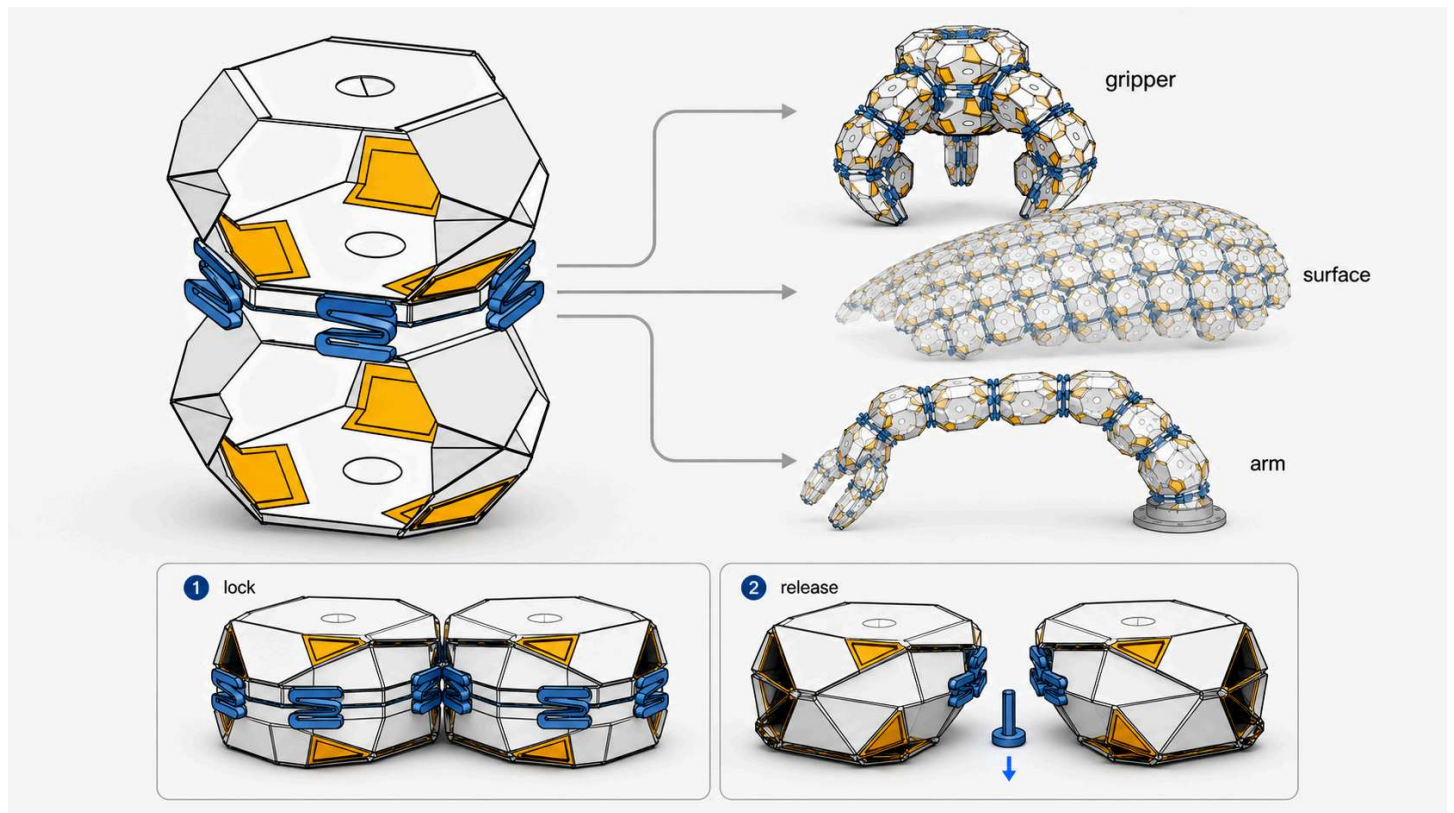


Lightweight Active Locking Interfaces for Origami-Based Modular Robots



Section: Robotics, Mechanical Engineering, Microengineering

Supervisors: Shaopeng Jiang, Ziqiao Wang, Prof. Jamie Paik

Number of Credits: 10 ECTS (300 hours)

Type: 60% mechanical design and testing, 20% Control, 20% Evaluation

Requirements: Mechanical design and prototyping, CAD modeling, and basic experimental analysis skills. Experience with pneumatics or robot control is a plus.

Description: To enable origami-based modular robots to form scalable reconfigurable systems, we aim to develop a lightweight active locking interface that allows modular robots to dock, lock, carry load, and release on command. Our lab works with origami-based 3DoF parallel robots that can fold flat, extend, and provide controlled roll, pitch, and heave motions. Existing projects already use these modules as grippers, serial robotic arms, and variable-shape surfaces. This project investigates how such modules can be mechanically connected through a compact locking interface that is compatible with identical docking faces, tolerant to realistic alignment errors, and suitable for hands-on prototyping. We are especially interested in concepts that combine lightweight pneumatic actuation with origami-integrated structures to build the active locking interface.

The project is intentionally open-ended. As a semester project, the student will propose, compare, prototype, and evaluate active locking concepts under the practical constraints of lightweight robot hardware. Through this project, the student is expected to develop expertise in modular origami robot design, fabrication, experimental evaluation, and scientific paper writing. The lab will provide access to plug-and-play pneumatic modular robot hardware, as well as training in pneumatic actuator design, hardware fabrication, and experimental testing of the locking mechanism.

Expected Work:

- Review modular robot docking, mechanical locking, self-alignment, lightweight mechanisms, and reversible release concepts.
- Analyze the locking requirements (size, weight, actuation constraints, etc.) of the existing 3DoF origami robot modules in the lab, ensuring compatibility with deployment and reconfiguration processes.
- Design and fabricate a selected locking interface using CAD and rapid prototyping methods.
- Establish the evaluation metrics for the prototype under the following conditions: docking, locking, load-bearing and release.

Contact: shaopeng.jiang@epfl.ch, ziquiao.wang@epfl.ch